Java ArrayList — Interview Notes

Beginner-friendly • Telugu meaning • Examples • Quick revision

1. What is ArrayList?

Definition: ArrayList is a class in the Java Collections Framework that provides a **resizable array** for storing elements.
Simple Telugu: Multiple values ni store cheyadaniki use chestham. Size fixed ga undadu; elements add or remove chesthe size change avuthundi.

2. Why use ArrayList?

• Dynamic size

• Easy add and remove operations

• Many useful built-in methods

• Insertion order is maintained

3. Array vs ArrayList

Array	ArrayList
Size fixed	Size dynamic/resizable
length property	size() method
Primitive values can be stored	Objects are stored; wrapper classes for primitives
Fewer built-in operations	Many collection methods

4. Creating an ArrayList

```
import java.util.ArrayList;

ArrayList<String> students = new ArrayList<>();
```

5. Adding elements — add()

```
students.add("Ravi");
students.add("Sita");
students.add("Kiran");
```

add() method tho elements ni list lo add chestham.

6. Accessing elements — get()

```
System.out.println(students.get(0));
```

Index 0 nunchi start avuthundi. get() tho element ni access chestham.

7. Updating — set()

```
students.set(1, "Anu");
```

Index 1 lo unna value replace avuthundi.

8. Removing — remove()

```
students.remove(0);
```

Index 0 lo unna element remove avuthundi.

9. Finding size — size()

```
System.out.println(students.size());
```

List lo enni elements unnayo telusukodaniki size() use chestham.

10. Complete Example

```
import java.util.ArrayList;

class Main {
public static void main(String[] args) {
ArrayList<String> students = new ArrayList<>();
students.add("Ravi");
students.add("Sita");
students.add("Kiran");
System.out.println(students);
}
}
```

11. Important methods

Method	Use
add()	Add element
get()	Access element
set()	Update element
remove()	Remove element
size()	Number of elements
contains()	Check whether an element exists
clear()	Remove all elements
isEmpty()	Check whether list is empty

12. Important facts

Duplicates: Yes, ArrayList allows duplicate values.
Null: Yes, ArrayList can contain null.
Order: Insertion order is maintained.
Primitives: Use wrapper classes such as Integer instead of int.

13. One-Line Interview Answer

“ArrayList is a resizable array implementation in Java that stores elements dynamically.”

14. Quick Memory Trick

Array = Fixed size
ArrayList = Flexible size

15. Practice Questions

- What is ArrayList?
- Why is ArrayList called resizable?
- Array vs ArrayList?
- What are add(), get(), set(), remove(), size()?
- Can ArrayList contain duplicates and null?
- Can ArrayList store primitive types directly?